

Hypothesis Testing in E-Commerce
A Practical Analysis of a Synthetic Sales Dataset Using R

Sidney J Reynaud, Jr.

National University

Practical Application

July 19, 2026

Abstract

This report presents a practical application of inferential hypothesis testing methods using a synthetic e-commerce transaction dataset ($n = 500$) constructed in R. The dataset comprises ten variables spanning customer demographics, regional identifiers, product categories, discount application status, order totals, session duration, and binary purchase completion outcomes. Three research hypotheses were formulated to address substantive business questions: (a) whether the application of a discount significantly affects mean order total, (b) whether purchase completion rates differ across geographic regions, and (c) whether session duration is associated with purchase completion. To evaluate these hypotheses, an independent samples t-test, a chi-square test of independence, and a one-tailed independent samples t-test were employed. All analyses were conducted in RStudio using the *tidyverse* and *car* packages. Given the synthetic and randomized nature of the dataset, results across all three tests were expected to yield non-significant findings, consistent with the absence of engineered relationships. This report demonstrates that rigorous hypothesis testing procedures, even when applied to null results, provide a replicable and transparent framework for evidence-based decision-making in e-commerce analytics contexts.

Keywords: hypothesis testing, e-commerce analytics, independent samples t-test, chi-square test, R programming, synthetic data

Introduction

Modern e-commerce environments generate substantial volumes of transactional and behavioral data, and organizations increasingly rely on statistical analysis to inform decisions related to discount strategies, regional marketing, and user-experience optimization. However,

observed differences in customer behavior may arise from sampling variability rather than true population-level effects (Field, 2018). Hypothesis testing provides a probabilistic framework for determining whether such differences are statistically meaningful.

This report applies three classical hypothesis tests to a synthetic dataset constructed in R (Navarro, 2019). Although the dataset lacks real-world correlational structure, the analytical workflow mirrors the procedures used in applied business analytics: defining hypotheses, selecting appropriate tests, checking assumptions, computing effect sizes, and interpreting results in context.

Dataset Description

The synthetic dataset contains 500 customer transactions, each represented as a single row. As noted in the original document, “the dataset comprises 500 observations ($n = 500$), with each row representing a single customer transaction.” Ten variables capture identifiers, demographics, behavioral metrics, and outcomes.

Categorical variables include region (North, South, East, West), product category (Electronics, Clothing, Home & Garden, Sports), and customer age group (18–25, 26–35, 36–50, 51+). Binary variables include discount application (Yes/No) and purchase completion (1/0). Continuous variables include order total, drawn from a uniform distribution ranging from \$15 to \$500, and session duration, drawn from a normal distribution with a mean of 25 minutes and a standard deviation of 10 minutes. Because variables were generated independently, no true relationships exist among them.

Hypothesis Formulation

Three hypotheses were developed to address business-relevant questions:

Hypothesis 1: Discount Application and Order Total

- **H0:** Mean order total does not differ between discounted and non-discounted transactions.
- **H1:** Mean order total differs between the two groups.

Hypothesis 2: Region and Purchase Completion

- **H0:** Purchase completion is independent of region.
- **H1:** Completion rates differ significantly across regions.

Hypothesis 3: Session Duration and Purchase Completion

- **H0:** Mean session duration does not differ between completers and abandoners.
- **H1:** Completers have longer sessions than abandoners.

Test Selection and Justification

Independent Samples t-Test (Hypothesis 1)

Order total is continuous, and discount status is binary, making the independent samples t-test appropriate (Field, 2018). Welch's t-test was used to ensure robustness to unequal variances. Cohen's d was computed to quantify effect size.

Chi-Square Test of Independence (Hypothesis 2)

Both region and purchase completion are categorical variables. The chi-square test evaluates whether observed frequencies differ from expected frequencies under independence (Agresti, 2013). Cramér's V was computed to assess association strength.

One-Tailed Independent Samples t-Test (Hypothesis 3)

Session duration is continuous, and purchase completion is binary. Because the hypothesis predicts a specific direction (longer sessions for completers), a one-tailed t-test was used (Navarro, 2019). Cohen's d was again computed.

Results

Hypothesis 1: Discount vs. Order Total

The t-test yielded a non-significant result ($p > .05$). Cohen's d was below 0.20, indicating negligible practical significance. Discounts did not meaningfully affect order totals.

Hypothesis 2: Region vs. Purchase Completion

The chi-square test was non-significant ($p > .05$). Cramér's V was below 0.10, indicating a negligible association. Completion rates were statistically uniform across regions.

Hypothesis 3: Session Duration vs. Completion

The one-tailed t-test showed no significant difference ($p > .05$). Cohen's d again indicated a trivial effect. Session duration did not predict purchase completion.

Discussion

Although the dataset lacks real-world correlational structure, the analysis illustrates how hypothesis testing supports business decision-making. In applied settings, a non-significant discount effect might prompt reconsideration of broad discount strategies. Uniform regional completion rates could indicate that geographic targeting is unnecessary. A weak relationship between session duration and conversion might suggest the need for multivariate behavioral modeling.

Limitations include the synthetic nature of the data, absence of confounding variables, and lack of ecological validity. Real e-commerce datasets typically exhibit complex interactions that require multivariate approaches such as logistic regression or A/B testing (Kohavi et al., 2009).

Conclusion

This report demonstrates the application of three inferential tests independent samples t-test, chi-square test, and one-tailed t-test to a synthetic e-commerce dataset. Although all results were non-significant, the workflow reflects best practices in analytics: clear hypothesis formulation, appropriate test selection, assumption checking, effect size computation, and contextual interpretation. As e-commerce data grows in complexity, the ability to apply these methods rigorously remains essential for data-driven decision-making.

References

- Agresti, A. (2013).** *Categorical data analysis* (3rd ed.). Wiley.
- Field, A. (2018).** *Discovering statistics using R*. Sage Publications.
- Kohavi, R., Longbotham, R., Sommerfield, D., & Henne, R. M. (2009).** Controlled experiments on the web: Survey and practical guide. *Data Mining and Knowledge Discovery*, 18(1), 140–181.
- Navarro, D. (2019).** *Learning statistics with R: A tutorial for psychology students and other beginners*. University of Adelaide.